

Mirror Descent - Another View

$$\phi \text{ } \rho\text{-strongly convex}$$

$$\nabla\phi(\mathcal{D}) = \mathbb{R}^n$$

$$\min: f(x)$$

$$\text{st: } x \in \mathcal{X}$$

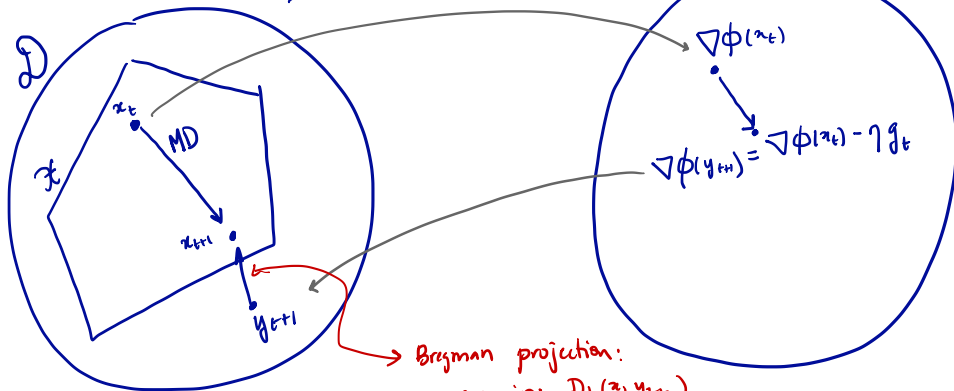
ϕ : "Mirror Map"

$\nabla\phi$ mapping to a "dual space".

Primal:

$\nabla\phi$

Dual:



→ Bregman projection:

$$x_{t+1} = \arg\min: D_\phi(x, y_{t+1})$$

$$\text{st: } x \in \mathcal{X}$$

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Connection to our example:

$$\phi(x) = \sum x_i \log x_i, \quad \mathcal{X} = \Delta = \{x \geq 0, \sum x_i = 1\}$$

$$\begin{aligned} \min: & f(x) \\ \text{s.t.} & x \in \mathcal{X} \end{aligned}$$

MD update:

$$x_t$$

$$y_{t+1} : \nabla \phi(y_{t+1}) = \nabla \phi(x_t) - \eta g_t, \quad g_t \in \partial f(x_t)$$

$$\hookrightarrow (y_{t+1})_i = (x_t)_i \cdot \exp \{-\eta (\nabla f(x_t))_i\}$$

$$x_{t+1} = \underset{\text{s.t. } x \in \Delta}{\operatorname{argmin}}: D_{\phi}(x, y_{t+1}) = \frac{y_{t+1}}{\|y_{t+1}\|_1}.$$